

Lecture 5: Functional Calculus

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Eigendecomposition: Square and Inverse of a Matrix

Consider a symmetric $d \times d$ matrix with eigenvalues λ_i and eigenvectors $\mathbf{u}_i \in \mathbb{R}^d$, and eigen-decomposition

$$\mathbf{A} = \sum_{i=1}^d \lambda_i \mathbf{u}_i \mathbf{u}_i^\top,$$

then we can simply compute the square or the inverse of $\mathbf{A}$ as follows:

$$\mathbf{A}^2 = \sum_{i=1}^d \lambda_i^2 \mathbf{u}_i \mathbf{u}_i^\top, \quad \mathbf{A}^{-1} = \sum_{i=1}^d \frac{1}{\lambda_i} \mathbf{u}_i \mathbf{u}_i^\top.$$

Definition 1 (Definition of matrix function)

Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $\mathbf{A} = \sum_{i=1}^d \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$, then

$$f(\mathbf{A}) := \sum_{i=1}^d f(\lambda_i) \mathbf{u}_i \mathbf{u}_i^\top.$$

Examples

1) Given a **polynomial** $p(\mathbf{A})$, we have

$$p(\mathbf{A}) = \sum_{i=1}^d p(\lambda_i) \mathbf{u}_i \mathbf{u}_i^\top.$$

For example, consider $p(\mathbf{A})$, where $p(x) = 1 + x + x^2$. We have

$$\begin{aligned} p(\mathbf{A}) &= \sum_{i=1}^d (1 + \lambda_i + \lambda_i^2) \mathbf{u}_i \mathbf{u}_i^\top \\ &= \sum_{i=1}^d \mathbf{u}_i \mathbf{u}_i^\top + \sum_{i=1}^d \lambda_i \mathbf{u}_i \mathbf{u}_i^\top + \sum_{i=1}^d \lambda_i^2 \mathbf{u}_i \mathbf{u}_i^\top \\ &= \mathbf{1} + \mathbf{A} + \mathbf{A}^2. \end{aligned}$$

Examples

2) Given a **matrix exponential** $e^{\mathbf{A}}$, we have:

$$e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} + \frac{1}{2}\mathbf{A}^2 + \frac{1}{3!}\mathbf{A}^3 + \dots$$

This can be shown using the Taylor expansion of the exponential function $e^x = 1 + x + \frac{1}{2}x^2 + \dots$, which yields

$$\begin{aligned} e^{\mathbf{A}} &= \sum_{i=1}^d e^{\lambda_i} \mathbf{u}_i \mathbf{u}_i^{\top} \\ &= \sum_{i=1}^d \mathbf{u}_i \mathbf{u}_i^{\top} + \sum_{i=1}^d \lambda_i \mathbf{u}_i \mathbf{u}_i^{\top} + \frac{1}{2} \sum_{i=1}^d \lambda_i^2 \mathbf{u}_i \mathbf{u}_i^{\top} + \dots \\ &= \mathbf{I} + \mathbf{A} + \frac{1}{2}\mathbf{A}^2 + \dots \end{aligned}$$

Loewner order

Definition 2 (PSD matrix)

A matrix $\mathbf{A}$ is positive semi-definite (PSD) if

$$\mathbf{u}^\top \mathbf{A} \mathbf{u} \geq 0 \text{ for all } \mathbf{u} \in \mathbb{R}^d,$$

or equivalently (if $\mathbf{A}$ symmetric) all the eigenvalues λ_i of $\mathbf{A}$ are non-negative, i.e.

$$\lambda_i(\mathbf{A}) \geq 0 \quad \forall i = 1, \dots, d.$$

Definition 3 (Loewner order)

Let $\mathbf{A}$ and $\mathbf{B}$ be two Hermitian matrices.

We say that $\mathbf{A} \succcurlyeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B}$ is positive semi-definite.

We will use the notation $\mathbf{A} \succcurlyeq 0$ to denote that $\mathbf{A}$ is PSD.

Proposition 1

If $\mathbf{X} \succcurlyeq \mathbf{Y}$ and $\mathbf{A}$ is a $d \times d$ matrix, then $\mathbf{A}^\top \mathbf{X} \mathbf{A} \succcurlyeq \mathbf{A}^\top \mathbf{Y} \mathbf{A}$.

Proof:

Without loss of generality, we assume that $\mathbf{Y} = 0$, then we need to prove that $\mathbf{A}^\top \mathbf{X} \mathbf{A} \succcurlyeq 0$. To do so, simply note that

$$\mathbf{u}^\top (\mathbf{A}^\top \mathbf{X} \mathbf{A}) \mathbf{u} = (\mathbf{A} \mathbf{u})^\top \mathbf{X} (\mathbf{A} \mathbf{u}) \geq 0,$$

where the inequality is due to $\mathbf{X} \succcurlyeq 0$.

□

Eigenvalues

We start with a theorem regarding the monotonicity of the eigenvalues, and a corollary when considering non-decreasing functions of matrices:

Theorem 1 (Eigenvalue monotonicity)

If $\mathbf{X} \preceq \mathbf{Y}$, then $\lambda_i(\mathbf{X}) \leq \lambda_i(\mathbf{Y})$ for all $i = 1, \dots, d$.

Corollary 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a non-decreasing function. If $\mathbf{X} \preceq \mathbf{Y}$, then $\text{tr}(f(\mathbf{X})) \leq \text{tr}(f(\mathbf{Y}))$.

(note that $\mathbf{X} \preceq \mathbf{Y}$ does not imply $f(\mathbf{X}) \preceq f(\mathbf{Y})$)

Proof corollary: It follows from the theorem and the definition of the trace:

$$\text{tr}(f)(\mathbf{X}) = \sum_{i=1}^d f(\lambda_i(\mathbf{X})).$$

Definition 4 (Monotonicity)

Functions of matrices are said to be monotone (increasing) if $\mathbf{A} \preceq \mathbf{B}$ implies $f(\mathbf{A}) \preceq f(\mathbf{B})$.

Theorem 2

The function $f(x) = \frac{1}{x}$ is matrix monotone (decreasing), i.e. if $\mathbf{X}$ is invertible and $0 \preceq \mathbf{X} \preceq \mathbf{Y}$, then

- a) $\mathbf{Y}$ is invertible*
- b) $\mathbf{X}^{-1} \succcurlyeq \mathbf{Y}^{-1} \succcurlyeq 0$*

Theorem Proof

a) Proving that $\mathbf{Y}$ is invertible:

Since $\mathbf{X}$ is assumed to be invertible, $\lambda_i(\mathbf{X}) \neq 0$. Therefore, we have $\lambda_i(\mathbf{X}) > 0$ for all i , which implies

$$\lambda_i(\mathbf{Y}) \geq \lambda_i(\mathbf{X}) > 0 \quad \forall i,$$

which in turns implies that $\mathbf{Y}$ is invertible.

Theorem Proof

b) Proving that $\mathbf{X}^{-1} \succcurlyeq \mathbf{Y}^{-1} \succcurlyeq 0$:

We first consider the special case where $\mathbf{Y} = \mathbf{I}$ (and we will then show how to reduce the proof to this special case). By assumption, we have $0 \preccurlyeq \mathbf{X} \preccurlyeq \mathbf{I}$, i.e.

$$0 \leq \lambda_i(\mathbf{X}) \leq 1 \implies \frac{1}{\lambda_i(\mathbf{X})} \geq 1,$$

which proves that

$$\mathbf{X}^{-1} \succcurlyeq \mathbf{I}.$$

Theorem Proof

$$\mathbf{X}^{-1} \succcurlyeq \mathbf{I}; \tag{1}$$

Let's now consider the general case where we assumed that $0 \preccurlyeq \mathbf{X} \preccurlyeq \mathbf{Y}$. By multiplying by $\mathbf{Y}^{-1/2}$ on both sides:

$$\begin{aligned} 0 &\preccurlyeq \mathbf{X} \preccurlyeq \mathbf{Y} \\ \implies 0 &\preccurlyeq \mathbf{Y}^{-1/2} \mathbf{X} \mathbf{Y}^{-1/2} \preccurlyeq \mathbf{I} \\ \stackrel{Eq. (1)}{\implies} &(\mathbf{Y}^{-1/2} \mathbf{X} \mathbf{Y}^{-1/2})^{-1} \succcurlyeq \mathbf{I} \\ \implies &\mathbf{Y}^{1/2} \mathbf{X}^{-1} \mathbf{Y}^{1/2} \succcurlyeq \mathbf{I} \\ \implies &\mathbf{X}^{-1} \succcurlyeq \mathbf{Y}^{-1}, \end{aligned}$$

where the last step is obtained by multiplying by $\mathbf{Y}^{-1/2}$ on both sides. □

The Logarithm is Matrix Monotone

Theorem 3

The function $f(x) = \log x$ is matrix monotone, i.e. if $\mathbf{X}$ is invertible and $0 \preccurlyeq \mathbf{X} \preccurlyeq \mathbf{Y}$, then $\log \mathbf{X} \preccurlyeq \log \mathbf{Y}$.

Theorem Proof

Based on the identity $\log x = \int_0^\infty \left(\frac{1}{1+t} - \frac{1}{x+t} \right) dt$, the matrix form of $\log \mathbf{X}$ is

$$\log \mathbf{X} = \int_0^\infty \left[(1+t)^{-1} \mathbf{I} - (\mathbf{X} + t\mathbf{I})^{-1} \right] dt.$$

Theorem Proof

To check the last formula,

$$\log \mathbf{X} = \int_0^\infty [(1+t)^{-1}\mathbf{I} - (\mathbf{X} + t\mathbf{I})^{-1}] dt,$$

note that

$$\log \lambda_i = \int_0^\infty \left(\frac{1}{1+t} - \frac{1}{\lambda_i + t} \right) dt,$$

therefore

$$\begin{aligned} \log \mathbf{X} &= \sum_{i=1}^d \log(\lambda_i) \mathbf{u}_i \mathbf{u}_i^\top = \int_0^\infty \sum_{i=1}^d \left(\frac{1}{1+t} - \frac{1}{\lambda_i + t} \right) \mathbf{u}_i \mathbf{u}_i^\top dt \\ &= \int_0^\infty \left(\frac{1}{1+t} \underbrace{\sum_{i=1}^d \mathbf{u}_i \mathbf{u}_i^\top}_{=\mathbf{I}} - \underbrace{\sum_{i=1}^d \frac{1}{\lambda_i + t} \mathbf{u}_i \mathbf{u}_i^\top}_{=(\mathbf{X} + t\mathbf{I})^{-1}} \right) dt. \end{aligned}$$

Theorem Proof

Hypothesis: $\mathbf{X} \preccurlyeq \mathbf{Y}$. Then

$$\begin{aligned}\mathbf{X} + t\mathbf{I} &\preccurlyeq \mathbf{Y} + t\mathbf{I} \\ \implies (\mathbf{X} + t\mathbf{I})^{-1} &\succcurlyeq (\mathbf{Y} + t\mathbf{I})^{-1} \\ \implies -(\mathbf{X} + t\mathbf{I})^{-1} &\preccurlyeq -(\mathbf{Y} + t\mathbf{I})^{-1} \\ \implies \log \mathbf{X} &\preccurlyeq \log \mathbf{Y},\end{aligned}$$

where the last inequality is due to the definition of $\log \mathbf{X}$ (check!).